Variant Calling with GATK

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Introduction

Variant calling turns aligned reads into a list of single-nucleotide variants (SNVs) and small insertions/deletions (indels). GATK's HaplotypeCaller is the de facto standard: it locally reassembles regions with evidence of variation and emits probabilistic genotype calls in VCF format.

Prerequisites

Sequence alignment, BAM format, reference genome indexing.

Theory

HaplotypeCaller proceeds in four stages: define **active regions** (loci with potential variation), perform **local de novo assembly** of haplotypes, **align reads to haplotypes** via pair-HMM, and **genotype** using likelihoods. For cohorts, each sample is first emitted as a per-sample GVCF, then joint-genotyped with `GenotypeGVCFs` – this preserves reference evidence and scales linearly with samples.

Downstream filtering is done with variant-quality score recalibration (VQSR) for whole-genome data or hard filters for smaller studies.

Assumptions

Reads are duplicate-marked, base-quality-recalibrated (BQSR), and aligned to a standard reference; ploidy is specified.

R Implementation

The GATK workflow runs on the command line (Java-based); R interacts with VCF output via `VariantAnnotation`.

```r
library(VariantAnnotation)

# Sample VCF from the package
vcf_file <- system.file("extdata", "chr22.vcf.gz", package = "VariantAnnotation")
vcf <- readVcf(vcf_file, "hg19")

vcf                                     # summary (SNPs, indels, samples)
head(rowRanges(vcf))                    # variant coordinates
head(info(vcf)[, c("AF", "DP")])        # population AF, depth
geno(vcf)$GT[1:5, 1:3]                  # genotype calls (sample x variant)
```

A representative GATK command (not run here):

```
gatk HaplotypeCaller -R ref.fa -I sample.bam -O sample.g.vcf.gz -ERC GVCF
gatk GenotypeGVCFs   -R ref.fa -V cohort.g.vcf.gz -O cohort.vcf.gz
```

Output & Results

A VCF / GVCF with per-variant coordinates, alleles, annotations, and per-sample genotypes, depths, and likelihoods.

Interpretation

"Joint genotyping with GATK HaplotypeCaller called 4.2M SNVs and 0.8M indels across the cohort; after VQSR with 99.9% sensitivity we retained 3.9M high-confidence SNVs."

Practical Tips

- Always BQSR before calling; skipping it inflates false positives, especially for low-frequency variants.
- Use GVCF workflow for multi-sample studies; `HaplotypeCaller` in direct VCF mode is only for single-sample work.
- `DeepVariant` is a competitive deep-learning alternative, producing similar quality without assembly.
- For somatic calls, use `Mutect2`, not HaplotypeCaller.
- Filter by depth, strand bias, and mapping quality (see `VariantFiltration`); many library artefacts look like rare variants.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat